

Zirconium-silicide coating on zircaloy-4 substrate for accident tolerance: Effects on oxidation resistance and boiling

Gi Cheol Lee^a, Hyunwoo Noh^a, Hwasung Yeom^b, Hangjin Jo^b, Tong Kyun Kim^c, Minrak Kim^a, Kumar Sridharan^b, Hyun Sun Park^{c,*}

^a Department of Mechanical Engineering, POSTECH, Pohang 790-784, Republic of Korea

^b Department of Engineering Physics, University of Wisconsin-Madison, WI 53706, USA

^c Division of Advanced Nuclear Engineering, POSTECH, Pohang 790-784, Republic of Korea

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ABSTRACT

In this research, the cladding substrate, Zircaloy-4 was coated with a 3- μm zirconium silicide (ZrSi_2). At 700 °C in air, the ZrSi_2 coating prevented thermal oxidation and thereby protected the underlying Zircaloy-4 substrate. In contrast, uncoated Zircaloy-4 surface developed an oxide layer that was a few micrometers thick. To investigate the effects of the coating on boiling characteristics, a pool-boiling experiment with saturated water, and a Leidenfrost experiment at low Weber number were conducted under atmospheric pressure. Thermal oxidation of the ZrSi_2 -coated surface did not affect critical heat flux or Leidenfrost temperature, but thermal oxidation of the uncoated Zircaloy-4 substrate affected its boiling parameters. A thin ZrSi_2 coating can be applied easily to a Zircaloy-4 substrate, and may be considered for use in nuclear reactors.

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1. Introduction

Zirconium (Zr) alloy fuel cladding is widely used in light water reactors (LWRs) because of its low thermal neutron absorption cross section, good mechanical properties, and corrosion resistance in normal operating conditions (Sabol et al., 1994). LWRs operate at high power density to achieve efficient and inexpensive power generation, but a loss of coolant accident (LOCA) scenario makes Zr-alloys very susceptible on severe oxidation and degradation (Zinkle et al., 2014). The prime interest in nuclear safety is to prevent radiation leakage and core melt-down accidents. Under accident conditions, losing coolant causes increase in the temperature of the reaction core, so fuel cladding is exposed to high-temperature water and steam. In such conditions, the Zr cladding reacts with water; the reaction produces hydrogen followed by cladding embrittlement (Rebak, 2015). The reaction between

water and Zr-alloy is exothermic; the heat that it produces can accelerate the progression of the accident (Terrani, 2018). To improve the reliability of nuclear safety systems, ATF (Accident Tolerant Fuel) designs have been developed not only to mitigate accident scenarios at high temperature but also to improve efficiency of fuel cycle (Kurata, 2016). The improved ATF cladding should have properties such as minimized core enthalpy input of hydrogen generation, maintenance of core coolability, and retention of fission products (Zinkle et al., 2014). Consequently, oxidation-resistant coatings have been evaluated as a short-term approach to increase oxidation resistance.

Transition metal silicides have superior oxidation resistance and mechanical properties, and have therefore been used as coating on structural materials in high-temperature applications (Berztsis et al. 1992). Zr silicides (ZrSi_x ; $x = 0.5, 1, 2$) have melting temperature greater than 1600 °C (Gueneau et al., 1994), low neutron penalty, and good oxidation resistance, and are therefore attractive candidates as ATF Cladding (Yeom et al., 2017). Furthermore, ZrSi_x has similar linear thermal expansion coefficients to Zr cladding substrate, and compositional compatibility with it. ZrSi_x prevents oxidation of underlying on Zircaloy-4 substrate without spallation or cracking (Yeom et al., 2018a). Before an ATF material can be used in practice, its thermal-hydraulics characteristics

Abbreviations: Zry-4, Zircaloy-4; OS Zry-4, Pre-oxidized Zircaloy-4; ZrSi_2 -Zry, ZrSi_2 coated Zircaloy-4; OS ZrSi_2 -Zry, Pre-oxidized ZrSi_2 coated Zircaloy-4; CHF, Critical Heat Flux; MFB, Minimum Film Boiling; FIB, Focused Ion Beam; SEM, Scanning Electron Microscope; EDS, Energy Dispersive x-ray Spectroscopy; AFM, Atomic Force Microscopy; PID, Proportional-Integral-Derivative.

* Corresponding author.

E-mail address: hejsunny@postech.ac.kr (H. Sun Park).

Nomenclature

Roman Symbols

c_p	heat capacity (J/K)
g	gravitational constant (m/s^2)
H	enthalpy (J/kg)
k	thermal conductivity ($\text{W/(m}\cdot\text{K)}$)
l	droplet diameter (m)
L	latent heat (kJ/kg)
m	mass (kg)
P	pressure (N/m^2)
q''	heat flux (kW/m^2)
R	surface roughness (nm)
t	time (s)
T	temperature ($^{\circ}\text{C}$)
v	impacting velocity of droplet (m/s)
x	distance (m)

Greek Symbols

β	receding contact angle ($^{\circ}$) or $(\rho c_p k)^{-1}$
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δ	oxide layer thickness (m)
η	dynamic viscosity ($\text{kg/(m}\cdot\text{s)}$)
ρ	density (kg/m^3)
σ	surface tension (N/m)
ϕ	Orientation of a heating surface ($^{\circ}$)

Subscripts

a	arithmetic average
E	evaporation
fg	phase change (liquid to vapor)
g	gas
i	interface
l	liquid
LFP	Leidenfrost
max	maximum
q	root mean square
s	surface

should be evaluated. Under nucleate boiling, the cladding surface can achieve efficient heat removal. However, after critical heat flux (CHF), a vapor film covers the cladding surface and causes rapid decrease in heat flux and increase in cladding temperature. Therefore the boiling transition, CHF, will be central to defining safety margins during normal operation and accident scenarios in nuclear reactors. The other important boiling transition point is minimum heat flux (MHF) representing boiling regime change from film boiling (post-CHF) to nucleate boiling. During a LOCA, this post-CHF regime can happen until an emergency core cooling system (ECCS) actively cools the system. The reactor core is reflooded with coolant water and heat flux recovers after the temperature reaches the MHF temperature (T_{MHF}) or the Leidenfrost temperature. Therefore, to assess the normal operating and transient accident conditions in LWRs, the CHF and T_{MHF} must be evaluated. Previous tests of ATF cladding candidates were mainly on coated stainless steel substrate (Kam et al., 2015; Seo et al., 2015, 2016; Son et al., 2017, 2018; Lee and Kim, 2017) rather than Zircaloy-4 substrate (Yeom et al., 2018a). To evaluate the applicability of ZrSi_x for this task, pool boiling CHF and post-CHF regimes such as Leidenfrost temperature (T_{LFP}) should be quantified.

In recent boiling studies, based on advanced surface modification techniques such as nanofluid or microelectromechanical systems, boiling experiments have been actively conducted (Ahn et al., 2010, 2012; Kim and Kim, 2009; Chu et al., 2012; Kim et al., 2015) and these studies report significant impact of surface characteristics on the boiling phenomena. One notable example is anodic oxidation which generates a thin oxide layer on a Zircaloy-4 cladding surface with micro/nano surface structural features. These changes in morphology of the surface cause increase in its CHF, T_{MHF} , and T_{LFP} (Ahn et al., 2010; Kang et al., 2016; Kim et al., 2014; Lee et al., 2017). Therefore, it is important to understand the role of surface characteristics of newly developed ATF cladding materials on heat transfer behavior. Given that ATF designs proposed are based on increased oxidation resistance, the data on the effects of changes in reaction kinetics in these new ATF surface on boiling performance must be collected.

This study is a preliminary assessment of the effect of zirconium silicide coatings on boiling parameters and oxidation resistance of Zircaloy-4 surfaces. We performed oxidation test at 700°C in air and the oxide layer was evaluated by Focused Ion Beam (FIB)

and Scanning Electron Microscopy (SEM) in conjunction with Energy Dispersive x-ray Spectroscopy (EDS). The surface characteristics were investigated with surface roughness and contact angle depending on oxidation. Pool-boiling experiments with saturated water, and Leidenfrost experiments using water droplet with low Weber number were conducted under atmospheric pressure and the effect of surface oxidation on CHF and T_{LFP} were investigated.

2. Materials and methods

2.1. Fabrication of zirconium-silicide coating and pre-oxidation

ZrSi_x (targets obtained from American Elements Inc.) coatings were deposited on Zircaloy-4 (Zry-4) disks (diameter 20 mm, thickness 1 mm, UNS-R60804, obtained from Shenzhen Sunrise Metal Industry Co. Ltd) by magnetron sputter deposition. The Zry-4 disks were prepared by mechanical grinding (progressively with #1000, #1500, #2000 silicon carbide abrasive papers) and chemical cleaning using acetone and ethanol, then rinsing with deionized water to remove surface contaminants. The deposition process was performed at room temperature, and with the stage rotating at 10 rpm. Details of coating procedures including plasma cleaning (500 W for 5 min in 1.6 Pa) were described previously (Yeom, et al. 2017, 2018a). Multiple depositions were conducted to yield a measured coating thickness of $3\text{ }\mu\text{m}$ (Section 3.1). To investigate the effect of oxidation on boiling parameters, the ZrSi_2 -coated Zry-4 and uncoated Zry-4 surface were pre-oxidized in ambient air with heating rate of 22.5°C/min to 700°C , holding at 700°C for 1 h, then cooling at $\sim 5.6^{\circ}\text{C/min}$ to room temperature.

Four test specimens: Zircaloy-4 (Zry-4), pre-oxidized Zry-4 (OS Zry-4), ZrSi_2 -coated Zry-4 (ZrSi-Zry), and pre-oxidized ZrSi_2 -Zry (OS ZrSi-Zry) were investigated in this study (Fig. 1). The surface structures and topography were investigated using a field-emission scanning electron microscope (FE-SEM; JSM-7800F Prime with dual EDS) and atomic force microscope (AFM; VEECO Dimension 3100). Energy Dispersive x-ray Spectroscopy (EDS) after pre-oxidation detected oxygen increases of $\sim 54\%$ on Zry-4 and $\sim 36\%$ on ZrSi-Zry.

The roughness of each surface was measured three times using a surface profiler (Alpha-Step IQ; KLA Tencor) (Table 1). The rough-

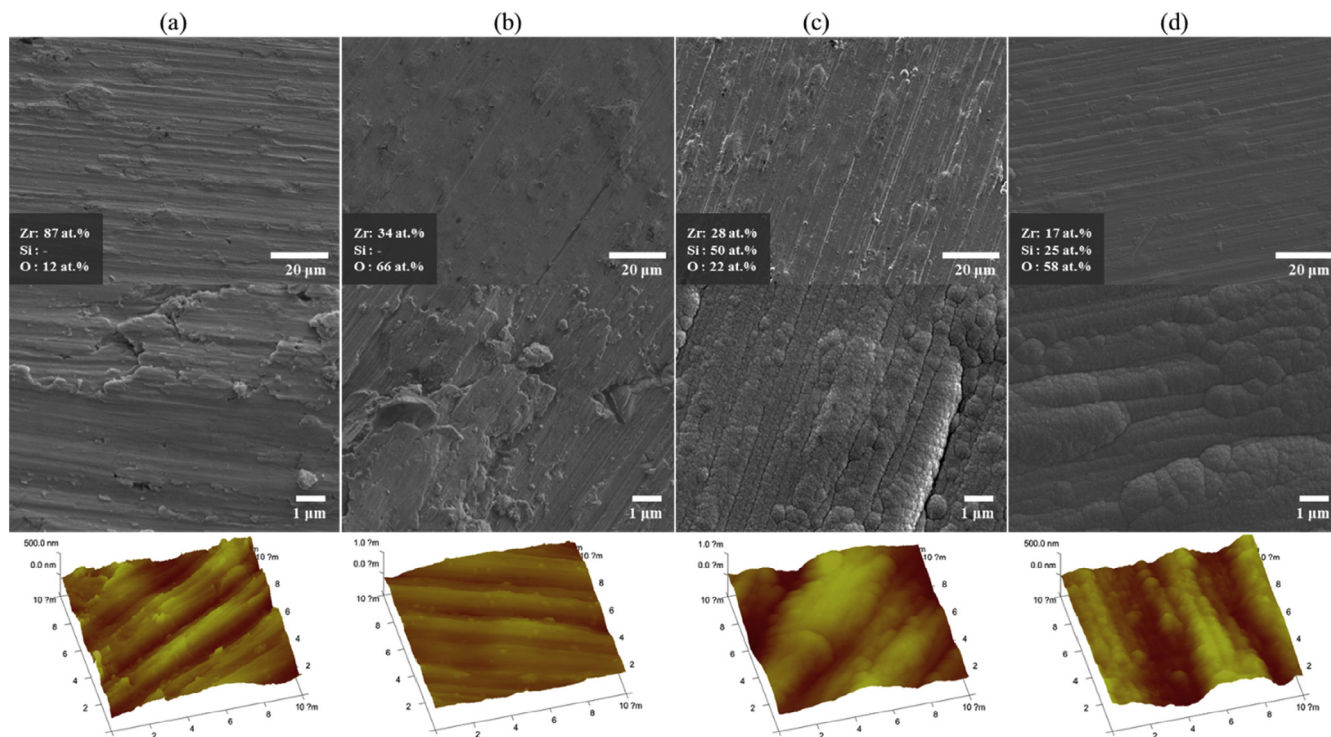


Fig. 1. Surface characteristics of the test specimens: (a) Zircaloy-4 (Zry-4), (b) pre-oxidized Zry-4 (OS Zry-4), (c) ZrSi₂-coated Zry-4 (ZrSi-Zry), (d) pre-oxidized ZrSi-Zry (OS ZrSi-Zry) by field-emission scanning electron microscopy with energy dispersive x-ray spectroscopy (top row), and atomic force microscopy (bottom row).

Table 1

Mean and standard deviations ($n = 3$) of surface roughness [nm] of test specimens.

Test specimen	Surface roughness [nm]					
	R_a		R_q		R_{max}	
	Mean	s.d.	Mean	s.d.	Mean	s.d.
Zry-4	153.7	13.8	202.3	10.5	1119.0	124.4
OS Zry-4	207.0	31.2	270.0	46.0	1492.7	246.7
ZrSi-Zry	187.7	11.5	241.7	15.3	1372.3	88.6
OS ZrSi-Zry	204.3	56.0	252.0	58.0	1230.7	256.7

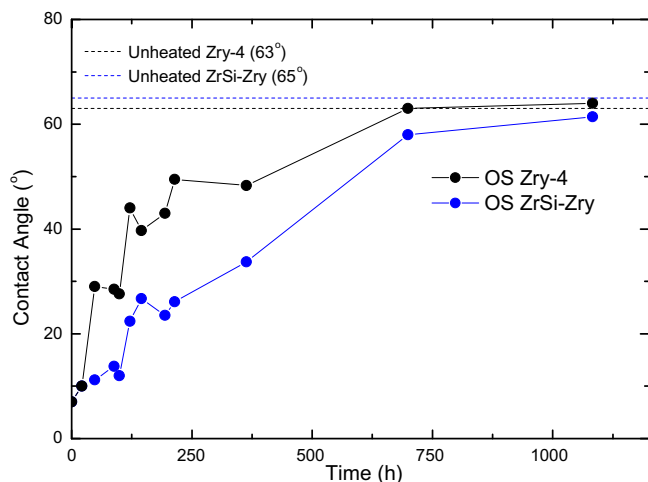


Fig. 2. Increase (recovery) in contact angle on pre-oxidized test specimen (OS Zry-4 and OS ZrSi-Zry). Results obtained under ambient conditions.

ness slightly increased after pre-oxidation. The contact angle was measured using SmartDrop (Femtofab) and 3- μ L drops of deionized (DI) water. The contact angle was not affected by the ZrSi₂

coating (about 63–65°). After pre-oxidation, the contact angle decreased, but it recovered in ambient conditions (Fig. 2). The temporary excitement of surface energy by thermal oxidation may be recovered as time goes by and this results in recovery to contact angle. Such hydrophobic recovery also appears on UV treated surfaces (Oláh et al., 2005; Antony et al., 2013; Caputo et al., 2008).

We also measured receding contact angle on Zry-4 and ZrSi-Zry surfaces by analyzing movement of liquid contact line during volume decrease of a droplet on surfaces because it is one of major factors on CHF. We posed a small droplet on the surfaces and drew it back using a syringe needle and a syringe pump. Contact line movement during droplet receding was recorded by a high speed camera and the angle between liquid and surface was measured during contact line change. As a result, receding contact angles for Zry-4 and ZrSi-Zry were 17.5° and 12.0°, respectively. For the OS surfaces, apparent contact angle was close to 0.0°, so we conclude that receding contact angle is about 0.0° on them because receding contact angle should be equal or smaller compared to the apparent contact angle.

2.2. Boiling experiment setup and procedures

2.2.1. Pool boiling

To measure the CHF of test specimen, a pool boiling facility was designed (Fig. 3a). The aluminum vessel (220 x 220 x 100 mm) was

filled with DI water. A reflux condenser condensed evaporated steam to maintain the water level inside the test pool. The test specimen was held by a polyetheretherketone jig and connected to a copper rod that delivers heat from cartridge heaters in the heat block. Six cartridge heaters with total capacity of 6 kW heat the copper rod; the generated heat was conducted to the test specimen. DI water was heated by an immersed heater that uses a proportional-integral-differential (PID) controller. Before each test, the DI water was de-gassed for 2 h. During the experiment, the DI water inside the pool was maintained at saturation temperature of 100 °C. Applied heat flux was calculated using Fourier's conduction equation ($q'' = k\Delta T/\Delta x$) from temperature measured by 0.5 mm K-type thermocouples (T/Cs) installed at 10-mm intervals. In this study, we defined CHF point as a point where surface temperature increases suddenly up to higher than 250 °C. At CHF, heat transfer is significantly deteriorated by the formation of vapor layer on the surface, which act as an insulation layer. The uncertainty of heat flux measurement was calculated as described earlier (Ahn et al., 2010), and the result represented as smaller than 50 kW/m².

2.2.2. Leidenfrost temperature

To define T_{LFP} , we measured the evaporation time t_E of an impinging droplet on heated test surfaces as a function of the surface temperature T_S . The Leidenfrost phenomenon stands for film boiling state of liquid droplet on heated surface. Therefore T_{LFP} is defined as T_S with lowest thermal efficiency; longest evaporation time (Bernardin and Mudawar, 1999, Kim et al., 2011, 2014). The temperature of the test specimen was maintained by a heater (Meivac) and a PID control system. Heat was transferred to the test specimen by conduction. To minimize the temperature difference between the test specimen and the heater, a connection jig was used to press the test specimen to the heater. We inserted a 0.5 mm K-type thermocouple in center of test specimen to measure its temperature. To generate drops of uniform size, liquid water was transferred from a syringe to a needle (31G) using a syringe pump at a constant rate. A water droplet with a diameter of 2 mm was dropped on the test surface from a 14-mm height. The corresponding Weber number $We = \rho v^2 l / \sigma = 7.6$. t_E of the drop

was measured at least three times for each T_S , and these repeated tests, the uncertainty was ~ 0.5 s. The uncertainty in temperature measurements was ± 1 °C.

3. Results and discussion

3.1. Oxidation behavior of coated and uncoated samples

To investigate the oxidation resistance of the test specimen, cross sectional image was obtained using EDS analysis. After pre-oxidation, the test specimen was sectioned using a precision cut-off and grinding machine (Accutom-50, Struers). The cross sections were polished progressively with silicon carbide abrasive papers (#600, #1200, #1500, #2000) for accurate measurement of oxide layer thickness. The oxide layer was distinguished by SEM imaging and EDS line-scan for oxygen (Fig. 4a, b). The outermost surface of the Zircaloy-4 reacted with oxygen to form an oxide layer with thickness of about 2.8–5 μm , which is consistent with previous oxidation results on Zircaloy-4 in similar oxidation conditions (Yeom et al., 2018b).

The ZrSi_2 coated layer was observed by imaging the cross-section of the samples and EDS mapping as a zone of silicon about 3 μm thick (Fig. 5) and this thickness was maintained after pre-oxidation. The detection of oxygen in the outermost region of ZrSi_2 coating indicates an oxide layer with around 0.3 μm thick formed on the coating's surface.

The cross-sectional observation via mechanical sectioning could affect microstructure in the oxide scale (e.g., pore structure or cracks) during polishing of the vertically-cut section. Therefore, focused ion beam (FIB; Helios, FEI) measurement was conducted for accurate characterization. To prevent surface damage, a protective platinum layer was deposited before FIB processing. The ion beam excavated some pits that were ~ 20 μm wide (Fig. 6) and the oxide layer was clearly distinguishable and its thickness was consistent with previous measurements. OS Zry-4 showed a pore structure within the oxide layer. OS ZrSi-Zry developed an oxide layer about 0.3 μm thick, but did not exhibit pores or cracks. Thus

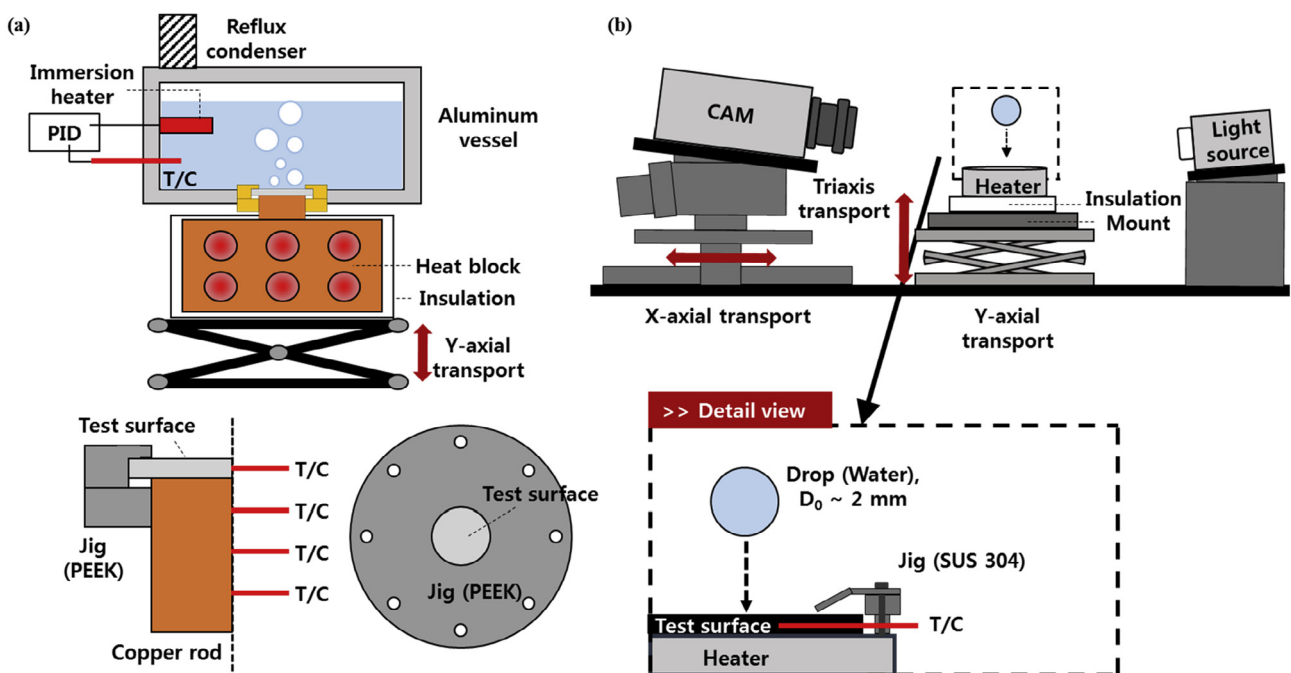


Fig. 3. Schematic illustration of the experimental facility for (a) pool boiling (b) Leidenfrost temperature measurements.

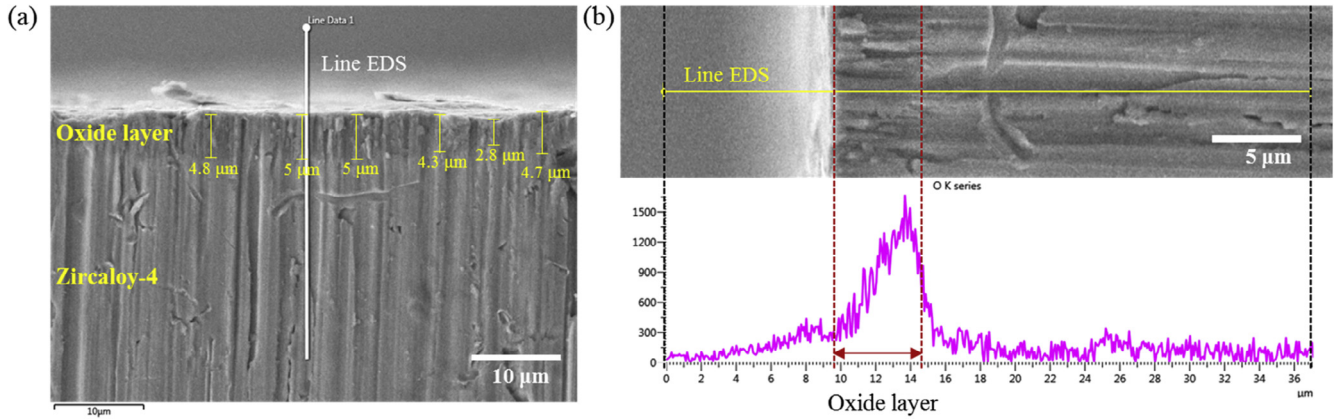


Fig. 4. Cross section view of pre-oxidized Zircaloy-4 (OS Zry-4) (a) SEM image, (b) oxygen concentration from EDS line-scan analysis, indicating the formation of an oxide layer about 5 μm in thickness.

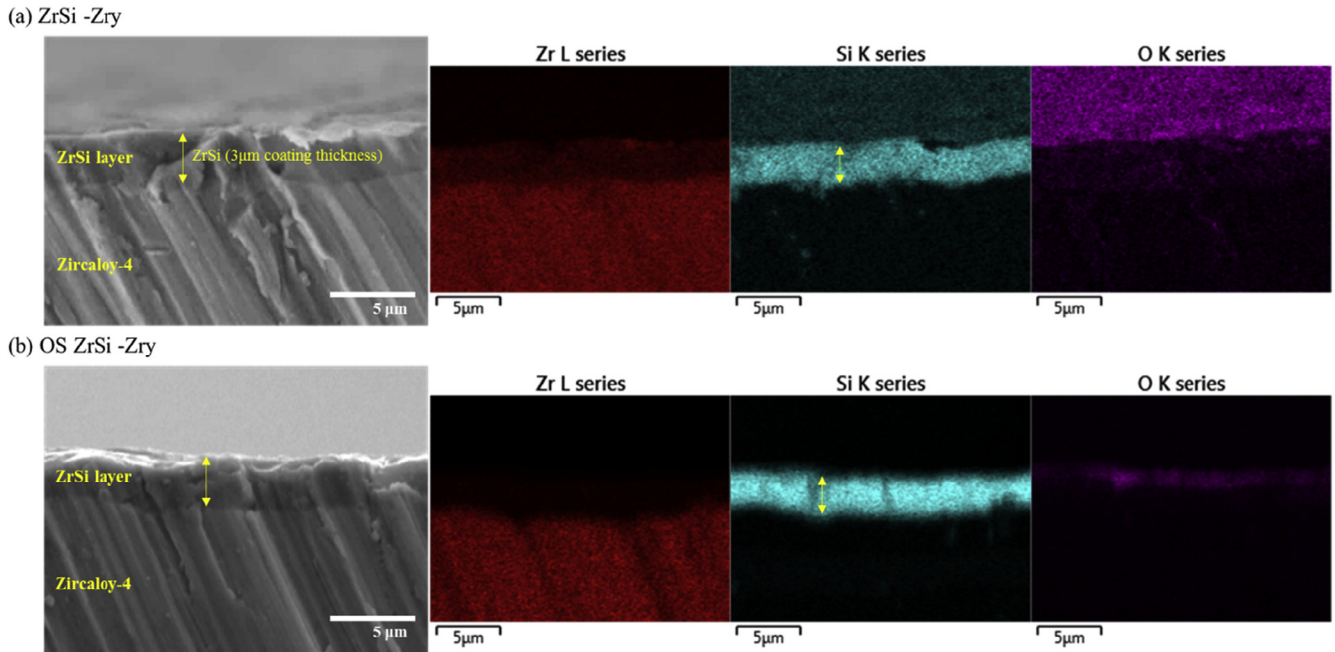


Fig. 5. Cross section view and EDS elemental mapping analysis (a) ZrSi_2 -coated Zry-4 (ZrSi-Zry) and (b) pre-oxidized ZrSi-Zry SEM image, showing a highly protective 3- μm thick ZrSi_2 coating layer with minimal surface oxidation.

the ZrSi_2 coating layer was largely preserved, and the underlying Zircaloy-4 substrate was protected from oxidation. EDS elemental mapping results (Supplemental Fig. 1) supported these results.

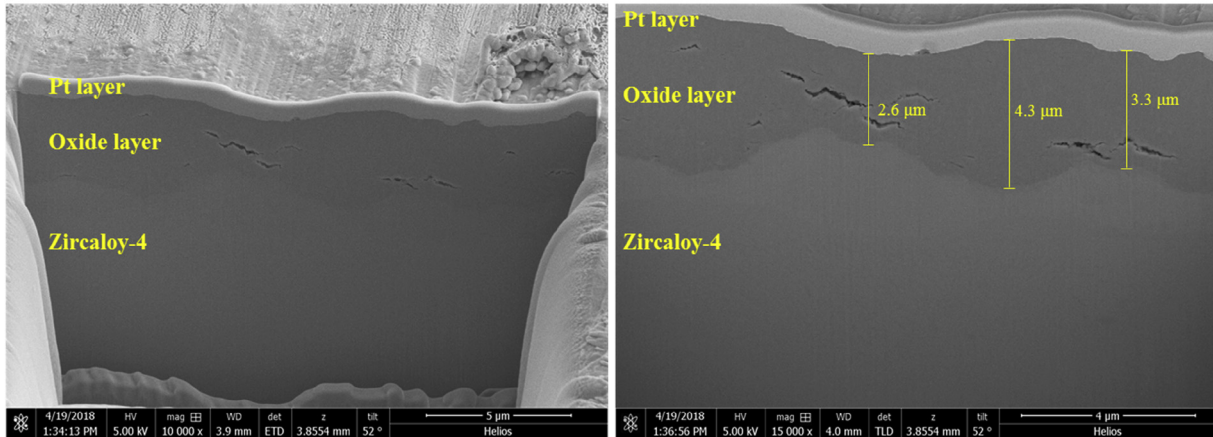
3.2. Pool boiling results (CHF)

When a bubble expands during boiling, there is a certain angle between the bubble interface and heating surface, which is called receding contact angle. Receding contact angle affects boiling behavior by changing bubble dynamics such as nucleation condition, bubble diameter, and bubble frequency (Jo et al., 2011). Kandlikar (2001) suggested a CHF model with respect to the receding contact angle on a surface as shown in Eq. (1). According to this model, CHF increases as receding contact angle decreases, representing a higher CHF on a more hydrophilic surface. However, for the surfaces used in this study, range of receding contact angle was 0.0–17.5°, which represented CHF variance smaller than 5% based on Eq. (1) (Fig. 7). Pool boiling experiments with saturated

DI water under atmospheric pressure yielded CHF of each specimen (Fig. 7). Average CHF was 858, 1004, 804, and 813 kW/m^2 for Zry-4, OS Zry-4, ZrSi-Zry, and OS ZrSi-Zry, respectively. CHF difference among Zry-4, ZrSi-Zry4, and OS ZrSi-Zry4 was about 50 kW/m^2 , which was similar with the uncertainty of CHF measurement, and it could be explained based on the Kandlikar's CHF model. However, OS Zry-4 represented about 17% higher CHF result than Zry-4, which is beyond the prediction from contact angle modeling. When a surface has superhydrophilic characteristic with capillary wicking, it can have higher CHF than that of Kandlikar's model (Ahn et al., 2010). However, we didn't observe capillary wicking on the experimental specimens used in this study. Therefore, we considered another surface factor, surface roughness, and it will be explained below.

$$q_c'' = h_{fg} \rho_g^{1/2} \left(\frac{1 + \cos \beta}{16} \right) \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \beta) \cos \phi \right]^{1/2} \left[\sigma g (\rho_l - \rho_g) \right]^{1/4} \quad (1)$$

(a) OS Zry-4



(b) OS ZrSi-Zry

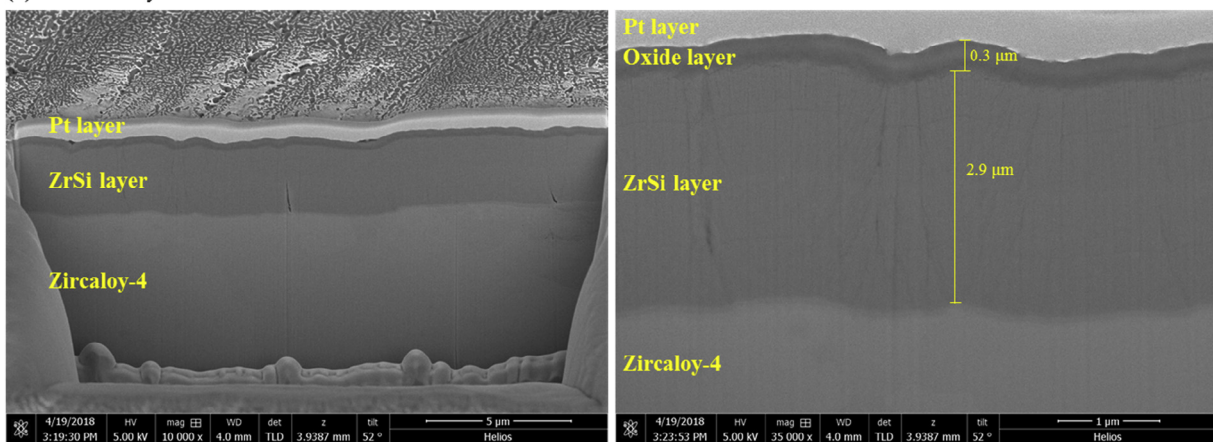


Fig. 6. Cross sectional view by Focused Ion Beam (FIB) prepared pre-oxidized test specimen (OS Zry-4 and OS ZrSi-Zry); Protective Pt layer is deposited before FIB sample preparation.

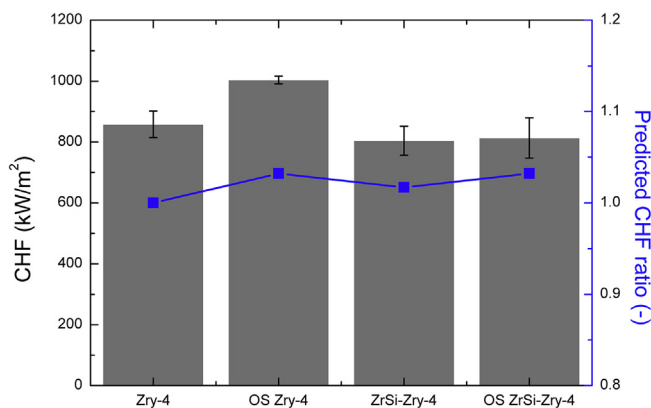


Fig. 7. Mean and standard deviation of measured CHFs, and predicted CHF ratio to the Zry-4 calculated by a previous CHF model (Kandlikar, 2001).

Surface roughness is known as one of the most influential surface parameters affecting CHF. Usually, CHF increases on modified surfaces that have high surface roughness, which was explained previously (Chu et al., 2012; Zou and Maroo, 2013; Seo et al., 2016; Son et al., 2017). Surface roughness may change CHF by affecting movement of bubble interfaces (Chu et al., 2012) or thin liquid layer beneath nucleated bubbles (Zou and Maroo, 2013). We measured surface roughness by AFM (Table 1), and roughness of

Zry-4 and ZrSi-Zry changed differently by oxidation effect. OS Zry-4 had about 30% higher average roughness R_a (207 nm), and maximum roughness R_{max} (1493 nm) than Zry-4 ($R_a = 154$ nm, $R_{max} = 1119$ nm) (Table 1). Roughness change during oxidation on Zry-4 specimens might have caused the CHF increase. However, on the other hand, ZrSi-Zry ($R_a = 188$ nm, $R_{max} = 1372$ nm) and OS ZrSi-Zry ($R_a = 204$ nm, $R_{max} = 1231$ nm) represented similar average and maximum roughness results with difference of smaller than 10% (Table 1). Superior material stability of $ZrSi_2$ coating protected surface change in high temperature condition with only small change of surface roughness. Because $ZrSi_2$ preserved its surface condition even in a harsh ambient, CHF change by oxidation was not significant on the $ZrSi_2$ specimens (Fig. 7).

Surface roughness may increase the effective contact line length around a bubble due to increased surface area. During bubble nucleation, there is a horizontal force balance between momentum-derived (expansion) force and surface-tension-derived (shrinkage) force. The direction of them is opposite, and CHF might occur when expansion force become larger than shrinkage force, as modeled previously (Kandlikar, 2001; Chu et al., 2012). Roughened surface provides longer contact line than flat surface, which increases shrinkage force by surface tension on the contact line. Therefore, high-roughness surface can maintain the horizontal force balance up to higher heat flux than flat surface, which corresponds to higher CHF result (Chu et al., 2012). In this study, average surface roughness increased after oxidation on the Zry-4 surface from 154 nm to 207 nm (Table 1). Therefore, we

can postulate that OS Zry-4 surface had higher CHF than Zry-4 because of increased horizontal shrinkage force by surface roughness.

Maximum surface roughness, which corresponds to the height of the structures, is also reported as significant surface factor to increase CHF (Zou and Maroo, 2013). Ridge-shaped structures with various heights were fabricated on Si and SiO₂ surfaces, to have $300 \leq R_{\max} \leq 1530$ nm. Pool boiling experiments with DI water revealed that CHF increased when maximum surface roughness exceeded a certain criterion and the increase amount remained constant for further increase of surface height. Zou and Maroo suggested that the CHF increased as a result of fragmentation and evaporation of non-evaporating film under a nucleated bubble. In their model, when the height of the structure was greater than the thickness of the non-evaporating film, the film could be broken, so CHF increased by evaporation of the fragmented liquid drops. They postulated the thicknesses of the non-evaporating films as 450 nm and 900 nm for Si and SiO₂ surfaces, respectively, which might be different from the surface material. In our experiments, based on the CHF results which changed only on the Zry-4 case after oxidation, we might conjecture that $R_{\max}$ of Zry-4 is between 1119 and 1493 nm and that of ZrSi-Zry is larger than 1372 nm.

3.3. Leidenfrost temperature (T_{LFP}) results

The evaporation time, t_E of a single droplet was measured to determine heat transfer rate and Leidenfrost temperature, T_{LFP} (Fig. 8a). t_E is defined as the time from the drop's contact with the heated surface until the drop evaporates completely. As the surface temperature, T_s increased, t_E initially decreased due to the accelerated nucleated boiling on the droplet, but when T_s exceeded a certain value, t_E started to increase (transition regime). Due to unstable dynamics during transition boiling, small parts of droplets came apart on overheated surface and underwent film boiling. This process results in increase and fluctuation of t_E (Bernardin and Mudawar, 1999; Lee et al., 2017, 2018). The trend of t_E was almost the same whether the ZrSi₂ coating was present or not. After pre-oxidation, both surfaces showed lower t_E trends, as a result of thermal resistance of oxidation layer. As mentioned in 2.2.2, T_{LFP} has been classically defined as T_s at which t_E is greatest. Based on this criterion, T_{LFP} was 300 °C on Zry-4, 354 °C on OS Zry-4, 345 °C on ZrSi-Zry, and 345 °C on OS ZrSi-Zry. The result on Zry-4 was consistent with data obtained previously under similar conditions (Kim et al., 2014) on the same material, with $We < 10$.

T_{LFP} was about 45 °C higher on ZrSi-Zry than on Zry-4. Results of the pool quenching experiment in atmospheric saturated water, increased T_{MFB} about 20 °C, were consistent with our results (Yeom et al., 2018a). The rewetting temperature (T_{MFB} or T_{LFP}) depends on the thermal properties of the surface (Kikuchi et al., 1985; Kang et al., 2018). The increased T_{MFB} was explained by reduction in thermal conductivity of sputtered coating process (Yeom et al., 2018a). This tendency of increased rewetting temperature was greater in the Leidenfrost experiment with a single droplet than in the pool-quenching experiment.

Surface roughness also has a strong influence on T_{LFP} . During film boiling, intermittent liquid-solid contact is promoted by surface features that pierce the vapor layer (Kim et al., 2011; Kwon et al., 2013; Yeom et al., 2018b). As the surface roughness increases, a thicker vapor layer, and hence a higher T_s , is required to keep the film in the boiling state (Bernardin and Mudawar, 1999). Therefore, increased surface roughness on Zry-4 surface by thermal oxidation might influence the T_{LFP} results. However, in this research R_s is in the 150–210 nm range and the vapor layer thickness is on the order of tens of micrometers (Biance et al., 2003; Burton et al., 2012; Lee et al., 2018), therefore intermittent liquid-solid contact may be precluded and the piercing mechanism seems unlikely.

The oxide layer would lower the liquid-solid interface temperature. (i.e. surface cooling effect). Consider a surface that has been heated to a high T_s . As a droplet approaches and touches the surface, a vapor layer begins to form under the droplet (Lee et al., 2018). Upon impact of the drop, the cool liquid water reduces T_s . At that moment, the liquid-solid interface temperature, T_i is determined by thermal effusivity $e = (\rho k c_p)^{1/2}$ and the initial temperatures of the liquid and the substrate (Eckert and Drake, 1972).

$$T_i = \frac{e_s T_s + e_l T_l}{e_s + e_l} \quad (2)$$

If we assume that T_i is equal at T_{LFP} on OS Zry-4 surface and Zry-4 surface,

$$T_i = \left(\frac{e_s T_{LFP} + e_l T_l}{e_s + e_l} \right)_{Zry-4} = \left(\frac{e_s T_{LFP} + e_l T_l}{e_s + e_l} \right)_{OS\ Zry-4} \quad (3)$$

$$T_{LFP, OS\ Zry-4} = (T_{LFP, Zry-4} + x_1 T_l) \cdot \left(\frac{1 + x_2}{1 + x_1} \right) - x_2 T_l \quad (4)$$

where $x_1 = e_{s, Zry-4}/e_l$, $x_2 = e_{s, OS\ Zry-4}/e_l$.

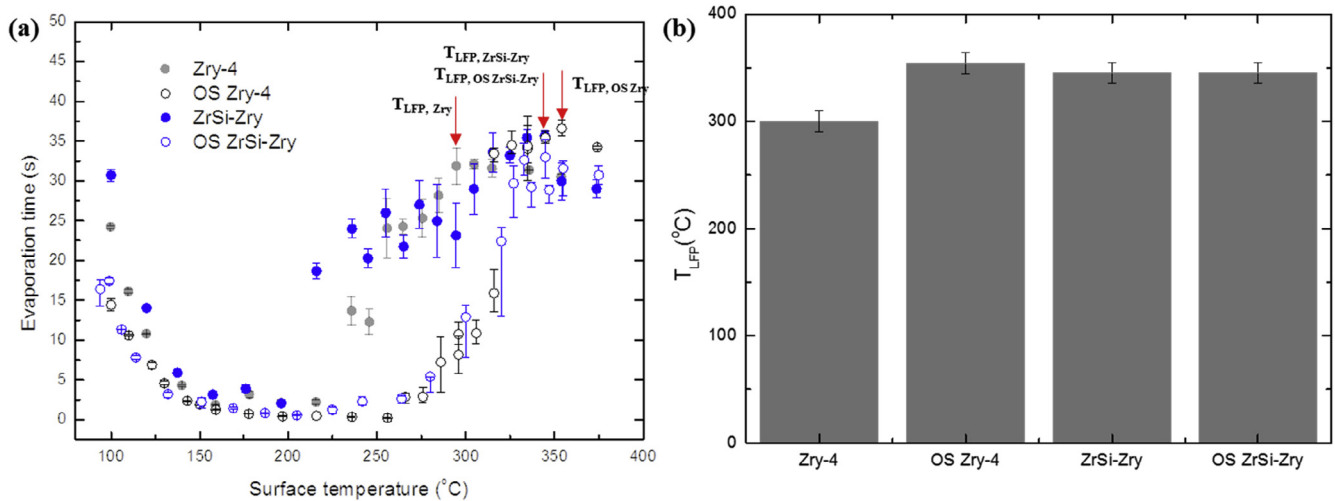


Fig. 8. Results for Leidenfrost experiment (a) Evaporation time as a function of surface temperature (b) Leidenfrost temperature (T_{LFP}) results on each test specimen.

Table 2
Thermal properties of Zry-4 and ZrO₂.

Thermal properties	Zry-4 (International Atomic Energy Agency, 2006)	ZrO ₂
Density (kg/m ³)	6510	5680 (Askeland and Wright, 2015)
Thermal conductivity (W/m/K)	15.4	2 (Nait-Ali et al., 2006)
Heat capacity (J/kg/K)	314.3	455 (Moriya and Navrotsky, 2006)

From Eq. (4), T_{LFP} on OS Zry-4 surface is calculated by 365 °C, which is about 11 °C higher than measured T_{LFP} result, 354 °C. We used the thermal properties of saturated liquid water, Zry-4 and ZrO₂ (Table 2), and measured T_{LFP} on Zry-4 for calculation.

The semi-empirical T_{LFP} model which considers transient conduction with a hot semi-infinite solid and thermal properties was also reported. (Baumeister and Simon, 1973). Depending on their correlation, we can acquire following relation between oxidized and un-oxidized surface when we assume that liquid-solid surface energy is same for both surfaces.

$$\frac{T_{LFP,OS\ Zry-4} - T_i}{T_{LFP,Zry-4} - T_i} = \frac{f(\beta)_{Zry-4}}{f(\beta)_{OS\ Zry-4}},$$

$$f(\beta) = \exp(3.066 \cdot 10^6 \beta) \operatorname{erfc}(1758 \sqrt{\beta}) \quad (5)$$

where $\beta = 1/\rho c_p k$.

From Eq. (5), T_{LFP} on OS Zry-4 surface is calculated by 390 °C, 36 °C higher than measured T_{LFP} results, 354 °C. The above predictions Eqs. (4) and (5) overestimate the increase of T_{LFP} by oxide layer because they does not consider the influence of oxide layer thickness. T_{LFP} increases as increase of oxide layer thickness by lowered T_i . In this study, the OS Zry-4 has relatively thin, about 5 μm thickness of oxide layer. Therefore increase of T_{LFP} should be smaller than above prediction equations. The experimental results and analytic model of T_{MFB} on oxidized Zircaloy-4 surface in saturated water pool quenching in atmospheric pressure were reported (Yeom et al., 2018b). Oxidized Zircaloy-4 with 5, 10, 25, 55 μm oxide layer thickness showed T_{MFB} results of 325, 355, 390, 430 °C. They analyzed instantaneous heat transfer during liquid-solid contacts in the stable film boiling on oxidized surface. It is hard to directly compare with OS Zry-4 result in this study and Yeom et al.'s results with uncertainties in thermal properties and the difference of physical phenomenon between pool quenching and droplet impingement. However, it implies that decreased e of the oxide surface decreases T_i and therefore would increase T_s required to achieve film boiling state.

4. Conclusion

The pool boiling CHF and T_{LFP} characteristics of zirconium silicide (ZrSi₂) coated on Zircaloy-4 has been investigated. The 3-μm thickness of ZrSi₂ coating was deposited by magnetron sputter deposition. The air oxidation resistance of the coating and the bare substrate were first analyzed using FIB, SEM, and EDS. The surface parameters of the oxidized and un-oxidized samples were measured and their effects on the results of CHF and T_{LFP} have been analyzed. The ZrSi₂ coating showed good oxidation resistance. In the oxidation test, uncoated Zircaloy-4 specimen developed an oxide layer with 2–5 μm thickness and internal pore structure. In the contrast, ZrSi₂ coating exhibited a surface oxide layer of 0.3 μm, and more importantly protected the underlying Zircaloy-4 substrate from oxidation. The coating itself did not influence the CHF results but, pre-oxidation of the samples affected CHF

results. Pre-oxidation increased CHF by about 17% on an uncoated Zircaloy-4 surface, but did not affect CHF of the ZrSi₂-coated surface. CHF increase by oxidation on Zry-4 surface was explained by increase of surface roughness after pre-oxidation. ZrSi₂ coating may reduce thermal conductivity and therefore increase T_{LFP} . Pre-oxidation increased the T_{LFP} on uncoated Zircaloy-4 surface, but not on the ZrSi₂-coated surface. The T_{LFP} increase on OS Zry-4 surface was explained by the thermal properties such as effusivity and liquid-solid interface temperature. Experimental results of CHF and T_{LFP} confirm that the surface parameters, their effective ranges, and major mechanisms that affect the boiling parameters are different for each boiling regime. In summary, ZrSi₂ coating was protective for oxidation. On ZrSi₂ coated surface, boiling characteristics (CHF and T_{LFP}) were constant after oxidation. Based on increased CHF and T_{LFP} results on OS Zry-4 surface, ATF cladding material for nuclear reactors with roughened and proper pre-oxidation will be beneficial from a heat transfer point of view.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.anucene.2018.11.019>.

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